Hilbert-Valued Characteristic Domination for $p \geq 2$

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Status

This packet gives a candidate partial result, likely valid, for Remark 12.10 of Yaroslavtsev's paper *Local characteristics and tangency of vector-valued martingales* [1]. It strengthens the earlier Hilbert $p = 2$ packet to every exponent $p \geq 2$. The result is still partial relative to the source problem, because it does not cover non-Hilbert UMD spaces or $1 \leq p < 2$.

Source Problem

The source is arXiv:1907.11588, PDF page 105, Remark 12.10. The source crop below states the open general-local-martingale issue and its discrete independent-increment reduction.

(12.11), (12.14), (2.9), and the fact that M and N are quasi-left continuous. $\square$

Remark 12.10. It is not known whether Theorem 12.8 holds for general local martingales. By Theorem 3.39 and by Proposition B.1 the main issue here is in proving a similar statement for discrete martingales with independent increments. Let us state this problem here as open. Let X be a Banach space, $1 \leq p < \infty$. Let $(\xi_n)_{n\geq 1}$ and $(\eta_n)_{n\geq 1}$ be X-valued mean-zero independent random variables such that for any Borel $B \in X \setminus \{0\}$

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B).$$

Does there exists a constant C (perhaps depending on p and X) such that

$$\mathbb{E}\Big\|\sum_n \xi_n\Big\|^p \leq C\mathbb{E}\Big\|\sum_n \eta_n\Big\|^p?$$

The next page adds that, by symmetrization, one can assume the variables are symmetric, and that even this case was not known to the author.

By a standard symmetrization trick [66, Lemma 6.3] one can assume that ξ_n's and η_n's are symmetric. But even the symmetric case is not known for the author.

In discrete form, the source asks whether independent mean-zero X-valued variables (ξ_n), (η_n) satisfying

$$\sum_n \mathbb{P}(\xi_n \in B) \le \sum_n \mathbb{P}(\eta_n \in B) \qquad (B \subset X \setminus \{0\} \text{ Borel})$$

must satisfy

$$\mathbb{E}\left\|\sum_n \xi_n\right\|^p \le C_{p,X}\mathbb{E}\left\|\sum_n \eta_n\right\|^p .$$

Idea of the Proof

The c_0 counterexample works because the norm sees the largest occupancy after the same aggregate jump mass is regrouped. Hilbert spaces do not have that freedom at exponents $p \ge 2$. A Hilbert-valued Rosenthal inequality says that the p-th moment of an independent mean-zero sum is equivalent to only two aggregate quantities:

$$\left(\sum_n \mathbb{E}\|Z_n\|^2\right)^{p/2} + \sum_n \mathbb{E}\|Z_n\|^p .$$

Both are monotone under the aggregate measure domination, by integrating $\|x\|^2$ and $\|x\|^p$. This proves the discrete comparison in Hilbert space for $p \ge 2$. The source paper already reduces the missing accessible-jump part of the general local martingale problem to precisely this discrete comparison; the continuous and quasi-left-continuous parts are already handled there.

Discrete Hilbert Comparison

Lemma 1 (Hilbert Rosenthal estimate)**.** *Let H be a Hilbert space, $p \ge 2$, and let $Z_1, \dots, Z_N$ be independent mean-zero H-valued random variables. Then*

$$\mathbb{E}\left\|\sum_{n=1}^N Z_n\right\|^p \eqsim_p \left(\sum_{n=1}^N \mathbb{E}\|Z_n\|^2\right)^{p/2} + \sum_{n=1}^N \mathbb{E}\|Z_n\|^p .$$

Proof. Let $S_k = \sum_{n=1}^k Z_n$. The Hilbert-valued discrete Burkholder-Davis-Gundy inequality gives

$$\mathbb{E}\sup_{1\le k\le N}\|S_k\|^p \eqsim_p \mathbb{E}\left(\sum_{n=1}^N \|Z_n\|^2\right)^{p/2};$$

this is the Hilbert case of the discrete estimate recalled in [2]. Since $(\|S_k\|)_k$ is a submartingale, Doob's L^p inequality also gives

$$\mathbb{E}\sup_{1\le k\le N}\|S_k\|^p \lesssim_p \mathbb{E}\|S_N\|^p ,$$

while the reverse inequality is immediate. Therefore

$$\mathbb{E}\|S_N\|^p \eqsim_p \mathbb{E}\left(\sum_{n=1}^N \|Z_n\|^2\right)^{p/2} .$$

Finally apply the scalar Rosenthal inequality to the independent nonnegative variables $Y_n = \|Z_n\|^2$ with exponent $p/2 \geq 1$:

$$\mathbb{E}\left(\sum_{n=1}^{N} Y_n\right)^{p/2} \eqsim_p \left(\sum_{n=1}^{N} \mathbb{E}Y_n\right)^{p/2} + \sum_{n=1}^{N} \mathbb{E}Y_n^{p/2}.$$

This is the displayed estimate. □

Proposition 1 (Aggregate domination in Hilbert spaces for $p \geq 2$)**.** *Let H be a Hilbert space and $p \geq 2$. Let $(\xi_n)_{n=1}^N$ and $(\eta_n)_{n=1}^N$ be independent mean-zero H-valued random variables. If*

$$\sum_{n=1}^{N} \mathbb{P}(\xi_n \in B) \leq \sum_{n=1}^{N} \mathbb{P}(\eta_n \in B) \qquad (B \subset H \setminus \{0\} \text{ Borel}),$$

then

$$\mathbb{E}\left\|\sum_{n=1}^{N} \xi_n\right\|^p \leq C_p \, \mathbb{E}\left\|\sum_{n=1}^{N} \eta_n\right\|^p.$$

Proof. For $q = 2$ and $q = p$, integrate $\|x\|^q$ against the two aggregate measures. The atom at 0 is irrelevant. We obtain

$$\sum_{n=1}^{N} \mathbb{E}\,\|\xi_n\|^q \leq \sum_{n=1}^{N} \mathbb{E}\,\|\eta_n\|^q, \qquad q \in \{2, p\}.$$

Applying Lemma 1 to (ξ_n), using these two inequalities, and applying Lemma 1 again to (η_n), gives the result. □

General Local Martingales

Theorem 1. *Let H be a separable Hilbert space and let $p \geq 2$. Let M and N be H-valued local martingales such that N is characteristically dominated by M in the sense of [1]. Then*

$$\mathbb{E}\sup_{t\geq 0} \|N_t\|^p \leq C_p \, \mathbb{E}\sup_{t\geq 0} \|M_t\|^p,$$

whenever the right hand side is finite.

Proof. We use the reduction developed in the source paper. The continuous part is handled by the characteristic-subordination/continuous domination estimate used in the proof of Theorem 12.8 of [1]. The quasi-left-continuous purely discontinuous part is exactly Theorem 12.8 of [1]. Thus, after the canonical decomposition, the only missing piece is the accessible-jump part.

For accessible jumps, use the accessible-jump decoupled tangent construction and the accessible-jump approximation proposition cited explicitly in Remark 12.10 of [1]. These approximate by martingales with finitely many predictable jumps and represent the corresponding decoupled tangent object, for almost every base point, as a sum of independent mean-zero increments. In Remark 12.10 the source states that the main remaining issue for general characteristic domination is precisely the discrete independent-increment aggregate-domination problem. Under characteristic domination, the total compensator of the accessible jumps of N is dominated by that of M; for the decoupled tangent sums this is exactly the aggregate domination hypothesis in Proposition 1. Applying that proposition pointwise to the decoupled accessible-jump sums gives the desired L^p comparison for the accessible parts. The standard tangent/decoupling estimate for Hilbert spaces, recalled as Theorem 1.1 in [1], transfers the estimate back to the original accessible-jump martingales, and the approximation then removes the finite-jump truncation.

Combining the continuous, quasi-left-continuous, and accessible estimates with the canonical-decomposition L^p bounds used in the source proof yields the displayed estimate. □

Verification Notes

The new proof step is Proposition 1. It is stable under symmetrization but does not need symmetry: independence and the mean-zero assumption are enough for the Hilbert Rosenthal estimate. The proof also explains why the c_0 counterexample cannot be transplanted into Hilbert space for $p \geq 2$: after Rosenthal, regrouping the same aggregate mass can change neither the total second moment nor the total p-th moment enough to create an unbounded ratio.

Novelty and Literature Check

The cheap run indexes were searched for the arXiv id and for the terms `characteristically dominated`, `aggregate domination`, `Hilbert`, `p>=2`, `Rosenthal`, and `Bichteler-Jacod`. The run already contained the broad c_0 counterexample and the Hilbert $p = 2$ packet, but no Hilbert $p \geq 2$ packet. Literature searches found Yaroslavtsev's BDG paper [2], Rosenthal/Pinelis moment tools [3], and Montgomery-Smith–Pruss comparison inequalities [4], but no exact recorded solution of the Hilbert $p \geq 2$ subcase of Remark 12.10.

Limitations and Next Routes

This packet does not solve $1 \leq p < 2$. The Rosenthal reduction used above is naturally aligned with $p \geq 2$, because the p-th moment controls both the quadratic characteristic and the large-jump characteristic. It also does not solve general non-Hilbert UMD spaces. A plausible next step is to test whether the same two-characteristic reduction holds in 2-smooth UMD spaces or in L^q-spaces with the Dirksen–Yaroslavtsev/Marinelli–Roeckner Burkholder-Novikov-Rosenthal estimates.

Human Review Recommendation

First verify Lemma 1, especially the use of scalar Rosenthal for $\sum \|Z_n\|^2$. Then verify that the source paper's accessible-jump reduction indeed needs only the discrete aggregate-domination comparison; this is the point explicitly isolated in Remark 12.10.

References

[1] I. S. Yaroslavtsev, *Local characteristics and tangency of vector-valued martingales*, arXiv:1907.11588, 2019.

[2] I. S. Yaroslavtsev, *Burkholder-Davis-Gundy inequalities in UMD Banach spaces*, arXiv:1807.05573, 2018.

[3] I. Pinelis, *Rosenthal-type inequalities for martingales in 2-smooth Banach spaces*, arXiv:1212.1912, 2012.

[4] S. J. Montgomery-Smith and A. R. Pruss, *A comparison inequality for sums of independent random variables*, J. Math. Anal. Appl. 254 (2001), no. 1, 35–42; arXiv:math/9811124.

[5] H. P. Rosenthal, *On the subspaces of L^p $(p > 2)$ spanned by sequences of independent random variables*, Israel J. Math. 8 (1970), 273–303.